

Problem-Solution Fit

Date	27 june 2026
Team ID	xxxxxx (Team: Jaswanth Malapareddy, Madasu Rushi Venkata Ganesh, Rajani Meesala, Mounika Sonti, Nehan Chowdary Muvva)
Project Name	Opti crop-Smart Agricultural Production Optimization Engine”
Maximum Marks	5 Marks

Problem-Solution Fit Canvas: Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] Small to medium-scale Farmers Agricultural Researchers Policymakers	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] Low budget for expensive software subscriptions. Reliance on basic smartphones, often with poor internet connectivity in rural areas. Limited technical/software literacy.	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Agronomist Consultations: (Pros: Highly accurate and personalized / Cons: Expensive, slow, not scalable). Generic Weather/Farming Apps: (Pros: Easy to use, cheap / Cons: Lacks hyper-localized Machine Learning crop predictions).
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] Suboptimal crop selection leading to low financial yields. Wasted resources (fertilizers/water) due to guesswork. Frequency: High (Impacts every planting season).	9. PROBLEM ROOT / CAUSE [RC] The relationship between soil nutrients (N, P, K, pH) and climate parameters (temperature, rainfall) is highly complex and non-linear. Manual or intuitive calculations simply cannot accurately account for all these variables at once.	7. BEHAVIOR + ITS INTENSITY [BE] Farmers traditionally rely on generational knowledge and peer advice rather than scientific data (High intensity). Resource planning is often reactive rather than proactive (Medium intensity).

3. TRIGGERS TO ACT [TR] Approaching sowing/planting season. Receiving new soil test results (N, P, K, pH). Experiencing a poor harvest or declining profits. Sudden climate or weather shifts.	10. YOUR SOLUTION [SL] OptiCrop: A smart, mobile-responsive Machine Learning platform that instantly analyzes NPK, pH, and climate data to provide precise, data-driven crop recommendations and environmental suitability assessments for farmers.	8. CHANNELS OF BEHAVIOR [CH] Local farming cooperatives, government extension offices, peer-to-peer networks. ONLINE: WhatsApp farming groups, agricultural YouTube tutorials, regional farming forums.ONLINE OFFLINE
4. EMOTIONS BEFORE / AFTER [EM] Anxious about yields, frustrated by unpredictable climate, uncertain about fertilizer investments. AFTER: Confident in data-backed decisions, relieved by optimized resource usage, financially empowered.		